Long-read best practices

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Long-read technology

- Sequencing method that produces reads > 10kb, sometimes even megabases.
- PacBio and Oxford Nanopore have developed technologies for accurate long-read capturing.
- Used for detecting structural variants, full-length transcripts/isoforms, genome assembly, etc.
- Long-read is generally less accurate and more expensive than short read, but technology is improving

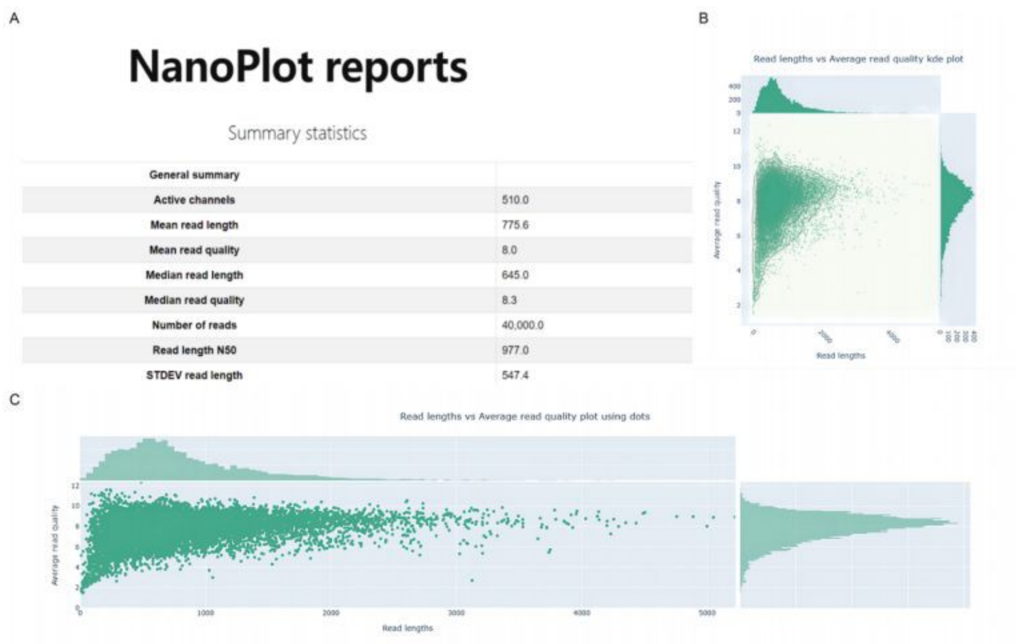
Oxford Nanopore vs PacBio

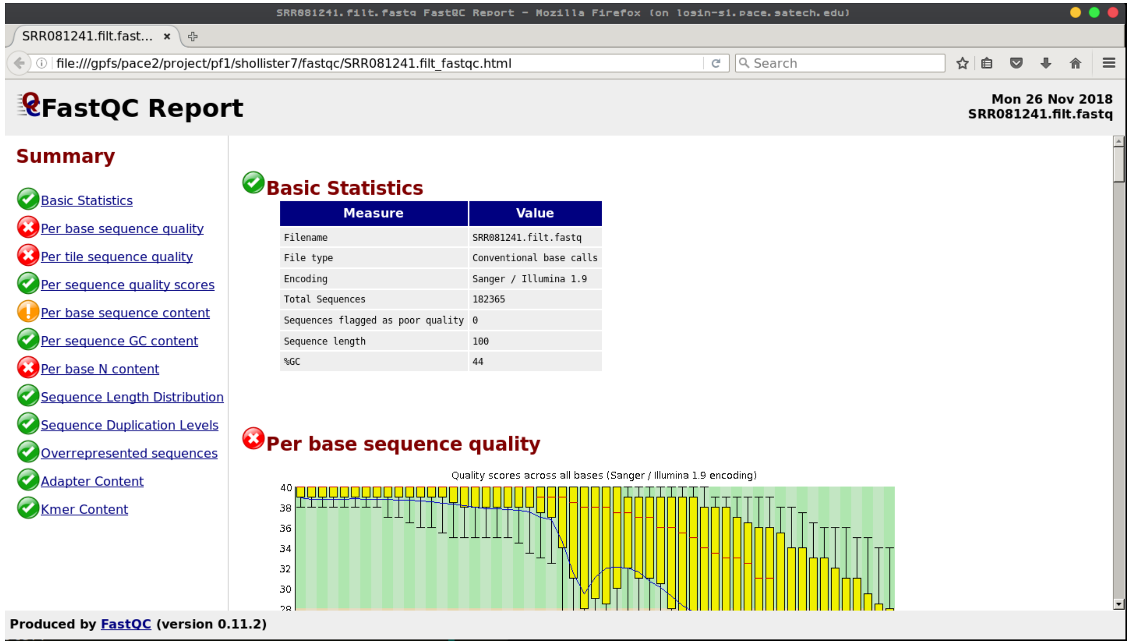
- Oxford Nanopore is used when you need very long reads
  - For example, structural variants are better studied using long reads
- PacBio is used when you need very high accuracy
  - SNP/INDEL calling, genome assembly

Feature	Oxford Nanopore	PacBio
Read Length	Very long (Megabases)	Long (>10kb)
Accuracy	90-98% accuracy	>99% accuracy
Sequencing Method	Real-time	Consensus (Read is read multiple times)
Cost per base	Lower, but generally more bases	Higher due to accuracy

Quality control

- Nanoplot – QC tool specifically made for long-read data.
- FastQC – can still be used for long-read data, but typically used for short-read data.





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Alignment for long-read RNA

- Minimap2 is dominant tool for long read RNA alignment.
- Benchmark studies mainly look at transcript level tools or assembly tools
  - Transcript assemblers, quantifiers, differential tools
- LoTempio et al – Benchamarks the alignment tools for long-read RNA
  - Minimap2 outperforms all of the other tools
  - Winnowmap2 was the fastest tool
- Alignment tool generally less important in long-read because the longer reads are often easier to align to the genome.

Dong et al. Benchmark study for long-read RNA analysis

- Isoform Reconstruction and Detection
  - StringTie2: Highest precision (predictions were correct)
  - Bambu: Highest power (more novel isoforms were detected)
  - FLAIR, TALON, cupcake all overpredicted – a lot of artifactual isoforms
- Differential transcript expression (DTE)
  - DESeq2, edgeR, limma all effective and performed similarly in differential transcript analysis
- Differential transcript usage (DTU)
  - DRIMSeq and DEXSeq were the most powerful DTU tools but had high false discovery rates; edgeR controlled FDR well but had the lowest power

Pardo-Palacios et al – Benchmark study for long-read transcript indentification

- All reads were aligned using minimap2
- Read quality and length, not quantity were the drivers of accurate transcript reconstruction
- Toos for transcript identification varied in performance
  - **Best Tools for reference based**: **Bambu** and **IsoQuant**, **FLAMES**.
  - **De Novo Transcript Detection – StringTie2** for assembly and **Isoquant** for quantification, **RNA-Bloom** and **rnaSPAdes** both prone to overcalling.

Dierckxsens et al – Benchmark study for long-read structural variant

- **cuteSV, SVIM, and pbsv** performed best overall, with pbsv being most accurate for SV position and length.
- **combiSV**, a new tool, combines outputs from 6 SV callers to improve recall, precision, and accuracy.
- Higher sequencing **depth doesn't always help**, but **longer reads** consistently improve SV detection.

Luth et al. – Genome assembly tools with long-read data

- **Flye** is the top-performing long-read assembler overall for PacBio CLR and ONT reads.
- **Hifiasm** and **LJA** are best for high-accuracy PacBio HiFi assemblies.
- **Miniasm**, **Raven**, and **wtdbg2** work well for small genomes with lower resource use.
- Longer reads improve reduce interruptions in large genomes but don't always enhance identity or gene recovery.

Varaint Callers for long-read DNA data

- Clair3
  - Deep learning-based
  - Top accuracy on ONT & PacBio HiFi
- PEPPER-Margin-DeepVariant
  - Ensemble model (PEPPER + DeepVariant)
  - Excellent for PacBio HiFi
- Mutserve2
  - Best for mitochondrial low-frequency variants
  - High sensitivity & F1 score

Differences from short-read

- Data comes in FAST5 format, not FASTQ.
  - Tools like Guppy and Bonito are used for base calling
- Trimming
  - Tools like FiltLong or NanoFilt are used to trim long-read data
- Deference in tools for variant calling and transcript identification/quantification
  - As discussed, different tools should be used when dealing with long-read data